

INGREDIENTS

Hamilton principle
 $\delta \mathcal{A}^\star[u, p, \underline{\Pi}] + \forall \delta p, \delta \underline{\Pi}, \begin{cases} \forall \delta \chi = 0 \text{ on } \partial\omega_t \\ \forall \delta \chi = 0 \text{ on } (\partial\omega_X)^{\text{Dirichlet}} \end{cases}$

& definition of the total energy
 $l^\star(\chi, p, \underline{\Pi}, T, \underline{C}_i^{-1}) = \hat{E}_{\text{tot}}^\star(\chi, p, \underline{\Pi}, T, \underline{C}_i^{-1}) - \rho T s^\star(\underline{\Pi}, T, \underline{C}_i^{-1})$

Thermal capacity

$$c = T \frac{\partial S}{\partial T} = \frac{\partial \hat{E}_{\text{tot}}}{\partial T}$$

Helmholtz free energy

$$\rho w^\star(T, \underline{\Pi}, \underline{C}_i^{-1}) = \rho T s^\star(T, \underline{\Pi}, \underline{C}_i^{-1}) - \rho \hat{e}_{\text{int}}(T, \underline{E}, \underline{C}_i^{-1})$$

$d/dt \downarrow$

Gibbs law

$$\rho \frac{dw^\star}{dt} = \rho T \frac{ds^\star}{dt} + \rho \frac{dT}{dt} s^\star - \rho \frac{d\hat{e}_{\text{int}}}{dt}$$

Fourier's law

$$\underline{Q} = \underline{K} \cdot \frac{\partial T}{\partial \underline{X}}$$

Viscoelastic stress-strain law

$$\frac{1}{2} \mathcal{L}_v(\underline{b}_e) \cdot \underline{b}_e^{-1} = -\underline{V}^{-1} : \underline{\tau}_{\text{NEQ}}$$

STATE LAWS

$$\frac{\partial \chi}{\partial \underline{X}} = -\frac{\partial \hat{E}_{\text{tot}}^\star}{\partial \underline{\Pi}} - \left(\frac{\partial s^\star}{\partial T} \right)^{-1} \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial T} \frac{\partial s^\star}{\partial \underline{\Pi}}$$

$$\frac{\partial p}{\partial t} = \text{DIV}_{\underline{X}}(\underline{\Pi}) - \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial \chi}$$

$$\frac{\partial \chi}{\partial t} = \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial p}$$

$\downarrow$

$$\left(\frac{\partial T}{\partial t} \right)_{\text{rev}} = \left(\frac{\partial s^\star}{\partial T} \right)^{-1} \frac{\partial s^\star}{\partial \underline{\Pi}} : \underline{\nabla}_{\underline{X}} \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial p} \underline{S}$$

$$\begin{aligned} \left(\frac{\partial T}{\partial t} \right)_{\text{irr}} = & -\frac{\partial}{\partial \underline{X}} \cdot \left(\underline{K} \left(\frac{T}{\rho c} \right)^2 \frac{\partial}{\partial \underline{X}} \left(\frac{\partial s^\star}{\partial T} \right) \right) \\ & + \frac{1}{(\rho c)^2} 4 \left(\frac{\partial \hat{e}_{\text{int}}^\star}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right) : T \underline{N} : \left(\frac{\partial \hat{e}_{\text{int}}^\star}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right) \frac{\partial S^\star}{\partial T} \\ & - \frac{1}{\rho c} 4 \left(\frac{\partial S^\star}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right) : T \underline{N} : \left(\frac{\partial \hat{e}_{\text{int}}^\star}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right) \end{aligned}$$

Lie derivative of stress tensors

$$\mathcal{L}_v \underline{S} = \dot{\underline{S}}, \mathcal{L}_v \underline{\Pi} = \underline{F} \mathcal{L}_v \underline{S}$$

$$\begin{aligned} & \int_{\omega_X} \left(\frac{\partial f}{\partial \chi} \cdot \frac{\partial \chi}{\partial t} + \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial t} - \frac{\partial f}{\partial \underline{F}} : \dot{\underline{F}} \right) d\omega_X \\ & = \int_{\omega_X} \left(\frac{\partial f}{\partial \chi} \cdot \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial p} - \frac{\partial f}{\partial p} \cdot \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial \chi} + \underline{\nabla}_{\underline{X}} \frac{\partial f}{\partial p} : \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial \underline{\Pi}} \underline{S} \right. \\ & \quad \left. - \underline{\nabla}_{\underline{X}} \frac{\partial f}{\partial p} : \frac{\partial s^\star}{\partial \underline{\Pi}} \underline{S} - \frac{\partial f}{\partial \underline{\Pi}} : \underline{\nabla}_{\underline{X}} \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial p} \underline{S} \right) d\omega_X \\ & \quad + \int_{\partial\omega_X} \frac{\partial f}{\partial p} \cdot (\underline{\Pi} \cdot \underline{N}) d(\partial\omega_X) \end{aligned}$$

$$\frac{\partial f}{\partial T} \frac{\partial T}{\partial t} = \frac{\partial f}{\partial T} \left(\frac{\partial T}{\partial t} \right)_{\text{rev}} + \frac{\partial f}{\partial T} \left(\frac{\partial T}{\partial t} \right)_{\text{irr}}$$

$$\frac{\partial f}{\partial \underline{C}_i^{-1}} : \frac{\partial \underline{C}_i^{-1}}{\partial t} = -4 \left[\frac{\partial w_{\text{NEQ}}}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right] : \left[\underline{F}^{-1} \cdot \left(\underline{F}^{-1} \cdot \underline{V}^{-1} \cdot \underline{F} \right)^\top \cdot \underline{F} \right] : \left[\frac{\partial f}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right]$$

EVOLUTION

DGBA BRACKET STRUCTURE

$$\frac{d\mathcal{F}}{dt} = \{\mathcal{F}, \hat{\mathcal{E}}_{\text{tot}}\} + (\mathcal{F}, \mathcal{S}) + \text{boundary terms}$$

$$\{\mathcal{F}, \hat{\mathcal{E}}_{\text{tot}}\} = \int_{\omega_X} \begin{pmatrix} \frac{\partial f}{\partial \chi} \\ \frac{\partial f}{\partial p} \\ \frac{\partial f}{\partial \underline{\Pi}} \\ \frac{\partial f}{\partial T} \\ \frac{\partial f}{\partial \underline{C}_i^{-1}} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & \underline{\nabla}_{\underline{X}} \underline{\square}_{\text{left}} \underline{S} & -\left(\frac{\partial s^\star}{\partial T} \right)^{-1} \underline{\nabla}_{\underline{X}} \underline{\square}_{\text{left}} : \frac{\partial s^\star}{\partial \underline{\Pi}} \underline{S} & 0 \\ 0 & -\underline{S} \underline{\nabla}_{\underline{X}} \underline{\square}_{\text{right}} & 0 & 0 & 0 \\ 0 & \left(\frac{\partial s^\star}{\partial T} \right)^{-1} \frac{\partial s^\star}{\partial \underline{\Pi}} : \underline{\nabla}_{\underline{X}} \underline{\square}_{\text{right}} \underline{S} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial \chi} \\ \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial p} \\ \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial \underline{\Pi}} \\ \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial T} \\ \frac{\partial \hat{E}_{\text{tot}}^\star}{\partial \underline{C}_i^{-1}} \end{pmatrix} d\omega_X$$

$$(\mathcal{F}, \mathcal{S}) = \int_{\omega_X} \begin{pmatrix} \frac{\partial f}{\partial \chi} \\ \frac{\partial f}{\partial p} \\ \frac{\partial f}{\partial \underline{\Pi}} \\ \frac{\partial f}{\partial T} \\ \frac{\partial f}{\partial \underline{C}_i^{-1}} \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{\rho} \frac{\partial \underline{\square}_{\text{left}}}{\partial \underline{X}} \cdot \underline{K} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial \underline{\square}_{\text{right}}}{\partial \underline{X}} + \frac{4}{c^2} \left(\frac{\partial \hat{e}_{\text{int}}^\star}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right) : T \underline{N} : \left(\frac{\partial \hat{e}_{\text{int}}^\star}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right) & -\frac{4}{c} \left(\underline{\square}_{\text{right}} \cdot \underline{C}_i^{-1} \right) : T \underline{N} : \left(\frac{\partial \hat{e}_{\text{int}}^\star}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right) & 0 \\ 0 & 0 & 0 & -\frac{4}{c} \left(\frac{\partial \hat{e}_{\text{int}}^\star}{\partial \underline{C}_i^{-1}} \cdot \underline{C}_i^{-1} \right) : T \underline{N} : \left(\underline{\square}_{\text{left}} \cdot \underline{C}_i^{-1} \right) & 4 \left(\underline{\square}_{\text{right}} \cdot \underline{C}_i^{-1} \right) : T \underline{N} : \left(\underline{\square}_{\text{left}} \cdot \underline{C}_i^{-1} \right) \end{pmatrix} \begin{pmatrix} \frac{\partial S}{\partial \chi} \\ \frac{\partial S}{\partial p} \\ \frac{\partial S}{\partial \underline{\Pi}} \\ \frac{\partial S}{\partial T} \\ \frac{\partial f S}{\partial \underline{C}_i^{-1}} \end{pmatrix} d\omega_X$$

$$\text{boundary terms} = \int_{\partial\omega_X} \frac{\partial f}{\partial p} \cdot (\underline{\Pi} \cdot \underline{N}) d(\partial\omega_X) - \int_{\partial\omega_X} \frac{1}{\rho} \frac{\partial f}{\partial T} \underline{K} \left(\frac{T}{c} \right)^2 \cdot \frac{\partial}{\partial \underline{X}} \left(\frac{\partial s}{\partial T} \right) \cdot \underline{N} d(\partial\omega_X)$$